

Soham Lodh

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SUMMARY

B.Tech CSE (AI & ML) undergraduate at KIIT (CGPA: 9.31) with hands-on experience building and deploying end-to-end AI/ML pipelines and scalable backend services to production. Shipped three ML-powered applications—from feature engineering and model selection through FastAPI inference serving—alongside event-driven backend systems handling concurrent workloads. Actively expanding into Generative AI, LLM application development, and cloud-native deployment, with foundational exposure to ML workflows on Google Cloud.

TECHNICAL SKILLS

Programming Languages: Python, JavaScript, TypeScript, Java, C
ML & AI: Scikit-learn, XGBoost, NumPy, Pandas, Optuna, SHAP, PyTorch
Backend & APIs: FastAPI, Node.js, Express.js, REST API Design, Prisma ORM
Frontend: React.js, Next.js, Redux Toolkit, Tailwind CSS
Databases: PostgreSQL, MongoDB, SQL
Cloud & Deployment: Vercel, Render
Tools & Platforms: Git, GitHub, Postman, Inngest
CS Fundamentals: Data Structures & Algorithms, OOP, DBMS, Operating Systems

EXPERIENCE

Technical Member – Web Development

Aug 2025 – Present

Coding Ninjas KIIT Chapter

Bhubaneswar, India

- Architected an event-driven platform (Next.js, Hono, Prisma, MongoDB) with a real-time leaderboard aggregating concurrent score streams; engineered the write pipeline with atomic aggregation to eliminate race conditions under high-concurrency load—directly applicable to monitoring and automation workflows at platform scale.
- Built a secure, server-side quiz platform with session tracking, timed execution, and randomized question delivery; diagnosed and resolved session edge cases surfacing only under concurrent user load by reproducing failure conditions from runtime logs.
- Designed a Project Management System enforcing four-tier role-based access control via middleware; authored API contracts and permission-logic documentation consumed by downstream contributors, demonstrating cross-functional collaboration and engineering best practices.

Open Source Contributor

Jul 2025 – Oct 2025

GirlScript Summer of Code

Remote

- Implemented four authentication modules (OTP login, JWT session management, Google OAuth, password recovery) with unit tests covering success and failure paths across all flows; changes reviewed, approved, and merged into the production main branch.
- Diagnosed production-level defects by reconstructing failure conditions from logs and submitted targeted fixes merged into the codebase, improving system reliability across affected flows.

PROJECTS

RiskLens – Production ML Inference Platform

Jun 2026

Python · Logistic Regression · Optuna · SHAP · FastAPI · React · Render · Vercel

Live: credit-default-risklens.vercel.app *GitHub:* github.com/Soham-Lodh/RiskLens

- Designed and implemented an 18-stage ML pipeline over 50,000 records from three relational data sources: enforced stratified splitting before EDA to prevent data leakage, engineered three ratio features from raw bureau data, and ran Bayesian hyperparameter optimisation (Optuna, 50 trials) achieving **AUC-ROC 0.9834** and **KS 85.91** on held-out data.
- Deployed a production inference service via FastAPI + Uvicorn on Render with a React frontend on Vercel; integrated SHAP explainability and a logit-space credit scoring algorithm mapping default probability to a 300–900 scale with four-tier institutional risk classification—demonstrating end-to-end ownership of an AI application from training through live serving.

CareerLens – Multi-Target AI Prediction Platform

Jun 2026

Python · XGBoost · Linear Regression · Optuna · FastAPI · React · Vercel

Live: career-helper-ml.vercel.app *GitHub:* github.com/Soham-Lodh/Career-Helper

- Built a multi-target ML system on 15,000 records (30 features) that simultaneously predicts burnout score (Linear Regression, **$R^2 = 0.564$**), career readiness score (tuned XGBoost Regressor), and counseling likelihood (tuned XGBoost Classifier); enforced a strict 70/30 split upstream of all preprocessing to guarantee zero data leakage across all three targets.

- Deployed all three models behind a single `/predict` FastAPI endpoint, with a multi-step React frontend on Vercel; tuned each model independently via Optuna TPE (75–100 trials, MedianPruner, 3–5-fold CV)—demonstrating scalable AI application architecture serving multiple ML models through a unified API.

Core Work – Multi-Tenant SaaS Backend Platform

Jun 2026

React · Redux Toolkit · Express · Prisma · PostgreSQL · Inngest · Clerk · Vercel

Live: core-work.vercel.app GitHub: github.com/Soham-Lodh/Core-Work

- Engineered an event-driven automation pipeline using Inngest to process seven Clerk webhook event types into durable background functions, propagating workspace state changes to the database.
- Architected a normalized 7-model PostgreSQL schema with cascading deletes and two-tier role-based access control across 10+ REST endpoints; built optimistic Redux state management to eliminate full-page refetches on mutations, reducing perceived latency for all CRUD operations.

Prescripto – Medical Appointment Platform

Oct 2025 – Jan 2026

MERN Stack, JWT, Cloudinary, Vercel

Live: prescripto-o5lf.vercel.app Admin & Doctor Panel: prescripto-admin-lyart.vercel.app

GitHub: github.com/Soham-Lodh/Prescripto

- Shipped a full-stack medical platform across two independently deployed frontends – a patient app and a unified Admin/Doctor panel – backed by a centralized Express REST API on MongoDB Atlas, live on Vercel.
- Designed role-based JWT authentication with separate login flows, protected route middleware, and permission-scoped dashboards for three account types; hardened the API with input validation, bcrypt password hashing, and rate limiting to prevent credential abuse.

EDUCATION

KIIT University <i>B.Tech, Computer Science Engineering (AI & ML)</i>	2024 – 2028 <i>CGPA: 9.31</i>
Aditya Academy Senior Secondary <i>Class XII (CBSE)</i>	2024 <i>85%</i>
Aditya Academy Senior Secondary <i>Class X (CBSE)</i>	2022 <i>85.66%</i>

CERTIFICATIONS

- Classify Images with TensorFlow on Google Cloud – Google (2025)
- DSA in Python – NPTEL (2025)
- Postman API Student Expert – Postman (2025)
- Master Machine Learning for Data Science & AI – Codebasics (2026)